



Teleconnections in spatial modelling

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ABSTRACT

In pedology, spatial context is relevant to soil-landscape systems on at least three different scales: i) the scale of quasi-local processes, which are independent of influence from the direct or wider neighborhood, ii) the scale of short-range processes for example on the local hillslope or catena, and iii) the scale of long-range processes, or teleconnected systems. We can represent the effects of teleconnections using existing tools and covariates, but we cannot easily infer or identify their controls, landscape processes or landscape units. We consider that an ability to identify the relevant controls in teleconnected systems would greatly improve pedological interpretation and understanding. Such understanding relates to the interaction of environmental factors and processes in the spatial context, which is relevant for environmental mapping generally. Here we show that teleconnected systems can be disassembled and interpreted using contextual modelling in such a way that the controls, i.e. the cause, can be localized in space. We present examples of how teleconnected systems can be deciphered. The methodology is based on the previously described ConMap approach in combination with Random Forest's measures of local feature importance. ConMap uses elevation differences, computed along multiple rays radiating out from a center grid cell, as predictors instead of complex surface derivatives or decomposed scales of a digital elevation model (DEM) or terrain attribute. Using synthetic and real-world data sets, we show how to identify and interpret teleconnections in soil environmental systems. In the synthetic example, elevation peaks are shown to produce larger values of soil properties, while, in contrast, a valley-mountain system is the main control of soil texture in the real-world example. Our analyses of teleconnected soil environmental systems illustrate that the stochastic component of the universal model of spatial variation is an integral but typically unresolved part of the deterministic component.

1. Introduction

There are two fundamental paradigms for digital soil mapping: the state factor equation (Jenny, 1941) and the universal model of spatial variation (Matheron, 1971). The state factor equation describes soil variation as the result of climate conditions (*cl*), relief (*r*), parent material (*p*), and organisms (*o*) over time (*t*). The equation proposed by Jenny (1941) is:

$$S = f(cl, o, r, p, t)$$

This widely used CLORPT approach appears to ignore spatial autocorrelation, i.e. the general fact that things that are close to each other are more similar than things that are further apart (Tobler, 1969).

In contrast, the universal model of spatial variation (Matheron, 1971) recognizes that all types of soil variations can be described and modelled in terms of a deterministic component and a stochastic component:

$$Z(s) = Z^*(s) + \varepsilon'(s) + \varepsilon'$$

where $Z(s)$ is the soil property value, $Z^*(s)$ is the deterministic part (i.e. CLORPT) of the model describing structural variation, $\varepsilon'(s)$ is the stochastic part of the model consisting of (apparently) random variation that may be spatially correlated and ε' is the spatially uncorrelated random noise component.

Recent studies have shown that the stochastic part $\varepsilon'(s)$, i.e. the spatial dependence, can often be modelled or described by contextual

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spatial information and is thus potentially an integral part of $Z^*(s)$ (Behrens et al., 2010, 2014, 2018a, 2018b). This is because covariate information is relevant at different scales and spatial wavelengths, leading to complex and nonlinear interactions that manifest themselves in deterministic and/or seemingly stochastic parts of spatial dependence. Within the modelling, the stochastic part can be analyzed, e.g. using variograms. It can also be modelled directly as a structural component of machine learning models, if the set of corresponding covariates is available in the form of relevant contextual data. Therefore, $\varepsilon'(s)$ should only be treated independently from $Z^*(s)$ if a spatial autocorrelation is present in the residuals and appropriate covariate data is lacking, but not as a general component of the model.

Pedologically, the influence and importance of spatial context become clear when considering the three most important soil environmental response systems, which we define below.

1. **Local systems.** The most common application and general interpretation, of the CLORPT approach is based on the observation that under a certain general climate and vegetation regime and for a certain parent material a specific soil develops. This response system describes a quasi-local, one dimensional process acting down a vertical profile in the Z direction only with no lateral influences from surrounding neighborhoods, although this process changes over geographical space (Fig. 1).
2. **Catenary and short-range systems.** Also generally described within the context of CLORPT, but based on relatively short-range neighborhood influences, are gravity-driven hillslope or catenary response systems such as material translocations (Fig. 2). Catenary transport is a good example that $\varepsilon'(s)$ is actually a part of $Z^*(s)$. In a simplified catenary transport model, the material is eroded in steep and convex slopes, transported downhill and accumulated in gentler, concave foot slopes. This develops a spatially autocorrelated variation in soil thickness (and other related soil properties) along the catena, which is often effectively described by a few local or regional terrain covariates such as slope, slope position, flow accumulation and curvature. Another example for short range effects is the influence of shading effects of trees or hills on soil or animal habitats, although they might also be regarded as teleconnection effects. These can be taken into account by a hillshade model as predictor. The development of floodplains is also a short-range process that influences the soil properties. When a river is flooded, first the heavy part of the river load, such as pebbles and sand, is sedimented, followed by finer materials such as silt and finally clay. This process leads to a strong spatial dependence, which correlates to the distance to the river and can therefore be modelled by including this specific distance information as a predictor (Schönbrodt et al., 2013). These examples reflect two- and three-dimensional processes arising in response to lateral influences from surrounding neighborhoods operating in X and Y directions, as well as in Z direction.

Local processes: Weathering

$$S = f(\text{parent material, climate, ...})$$

Cause and symptom occur at the same location

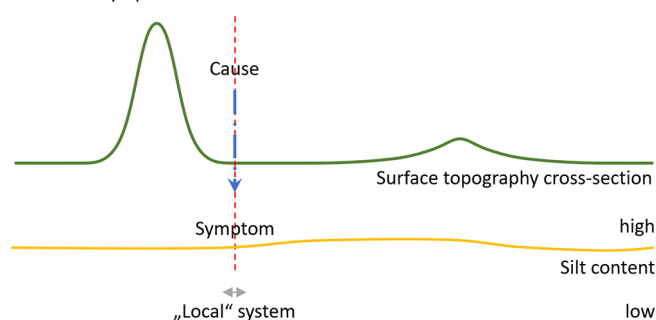


Fig. 1. Simplified model of a local response system.

Processes on a catena: Gravity induced material translocation

$$S = f(\text{slope, curvature, ...})$$

Cause and symptom are adjacent

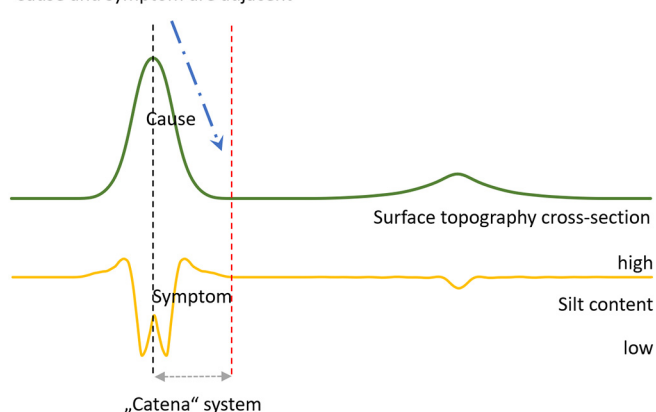


Fig. 2. Simplified model of a catenary response system.

3. **Teleconnected systems.** The third response system that we recognize is teleconnections, a concept that has not yet been much used in soil science. Teleconnections are known from atmospheric research and describe the interactions of environmental systems over longer distances (Trenberth et al., 1998), i.e. on regional, supra-regional, continental or global levels. Teleconnections arise from processes that operate over longer distances, which are initially unknown and influenced by the interactions of local site conditions and features or conditions located at some significant distance from the impacted site. One example in pedology is material removal and deposition by wind erosion as affected by contextual topographic controls (Fig. 3). Such a system cannot be detected using common GIS- and machine learning paradigms. The reason for this is that the interactions between e.g. surface shape and texture on the one hand and wind speed and precipitation on the other cannot be easily determined from available covariate data. Therefore, in such cases the stochastic variation $\varepsilon'(s)$ is often used as placeholder for the unknown (but actually deterministic) part. We consider it an advantage to reduce or remove $\varepsilon'(s)$ in predictive spatial models. The less spatial dependence that remains in model residuals, the better, and more completely, we are able to interpret the model outputs and the better we are able to understand the influence of the various environmental covariates on soil properties and soil formation (Behrens et al., 2014, 2018a, 2018b).

Teleconnections: Climate and the geomorphic signature

$$S = f(\text{interactions in the spatial context})$$

Cause and symptom lie far apart

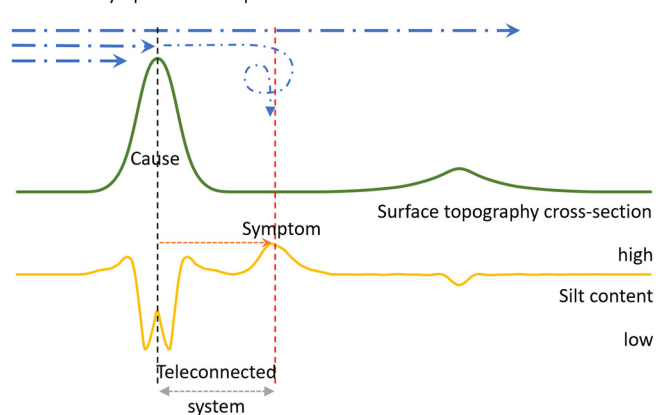


Fig. 3. Simplified model of a teleconnection response system.

The aims of this study are to show that the effect of teleconnections, i.e. the symptoms or spatial variability of soil properties, can be mapped using contextual modelling, and that teleconnected systems can be decomposed in such a way that the cause or control of that variability, can be spatially located and interpreted. Pedologically, this could be, for example, a remote peak, or a large water body, that influences the local climate pattern and therefore soil formation in a certain way, or an upwind ridge that encourages Aeolian deposition and accumulation. With existing methods and covariates, we may be able to map the effect of teleconnections, but we cannot identify, for example, a specific peak as the control of the observed soil spatial variation. The identification of specific environmental controls of soil properties in a spatial context provides a good basis for interpreting soil-environmental processes. We present examples of how teleconnected systems might be identified and deciphered. We develop and test the approach using one synthetic and one real-world dataset.

2. Material and methods

2.1. Modelling teleconnections

2.1.1. ConMap

Behrens et al. (2010) presented ConMap, a fundamentally different approach to terrain analysis. Instead of calculating conventional surface derivatives or decomposing the scales of a DEM or terrain attributes, the ConMap approach uses low-level analytical terrain information. ConMap is based on the differences in elevation from each pixel in a sparse circular neighborhood to the center pixel. These elevation differences (ConMap elevation differences, CED) are computed along multiple rays radiating out from a center grid cell and are directly used as predictors instead of complex surface derivatives of a digital elevation model (DEM) or terrain attribute. Fig. 4 shows an example of a mask of CEDs for one sample location of the real-world dataset. The data are stored in a table in $X, Y, Z_1, Z_2, \dots, Z_n$ format, where X and Y are the geographical coordinates and Z_1-Z_n are the columns for the

CEDs. Using sparse circular neighborhood masks, multiple scales can be computed and analyzed efficiently, describing a wide range of scales - from local to regional to supra-regional - without any accompanying decomposition of the scales, changes to the resolution of the DEM or increases in mathematical complexity.

The series of CEDs from rays of unrestricted distances that radiate out from a central grid cell in all directions (360 degrees) can be thought of as creating a unique representation of the location and spatial context of each point in space that has similarities to a spectral signature in soil spectroscopy, but in two dimensions. Every point on the earth will have a unique geomorphic signature (Pike, 1988) defined by the distances to, and elevations of, all surrounding points (Fig. 4). Locations that exhibit similar patterns with similar distances to points of similar elevation in similar directions are considered likely to exhibit similar environmental and topographic conditions and hence, likely to develop similar soil properties.

2.1.2. Random Forests

Random Forests (Breiman, 2001; Liaw and Wiener, 2019) are based on the well-known “classification and regression tree” (CART) approach introduced by Breiman et al. (1984). The ensemble is generated by averaging several trees based on different bootstrap sample sets. In addition, only a random subset of features is evaluated at each split. Random Forests has been used in various digital soil mapping studies over the past decade (Grimm et al., 2008; Behrens et al., 2010; Schmidt et al., 2014; Hengl et al., 2018) and for many other environmental problems (e.g. Cutler et al., 2007; Peters et al., 2007).

An advantage of the Random Forests implementation in R (Liaw and Wiener, 2019; R Core Team, 2019) is that it offers several possibilities to analyze the importance of features for a model. The overall importance of the feature as used in this study is determined by comparing the modelling error of the original data with a dataset in which a variable is randomly permuted. The increase in the prediction error is a measure of the importance of the permuted feature. This permits determination of which CEDs, are generally most important in predicting the specific attribute being modelled.

In contrast, the local or case-wise feature importance describes the importance of the i -th covariate for the j -th case or sample. This supports identification of which specific CEDs exert the greatest influence on the predicted soil property value at a single specific sample location. It is calculated as the average increase in residuals of a sample when a covariate is permuted and the sample is not part of the model (Liaw and Wiener, 2019).

2.1.3. Teleconnections

The spatial context, as computed using ConMap, can be used in combination with Random Forests' local or case-wise feature importance measurement to determine which local, regional or supra-regional landscape feature, or features, are most important for local soil formation and thus can be used to reveal the teleconnected system. The concept behind the approach is that different covariates are important in different regions of the data set, a concept that is related to non-stationarity.

Behrens et al. (2014) used the local feature importance to identify regions where a particular covariate most strongly affects soil formation. The idea presented here aims at the inverse. We want to elaborate teleconnection systems by determining the specific location(s) of the most influential CED features and thus the units of the geomorphic signature for specific sample locations at which a prediction is made. An advantage of the CEDs is that they provide exact relative coordinates calculated with respect to their location in a sparse neighborhood mask. Thus, the importance of each CED, that represents a covariate, can be mapped around any given sample. This allows the position of specific influential geomorphic features to be made directly identifiable when the local feature importance is visualized for the CEDs of a specific sample.

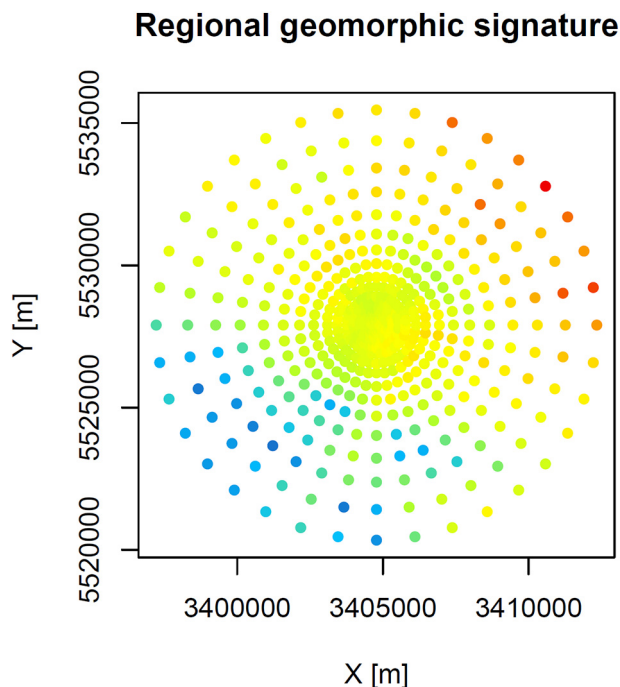


Fig. 4. Example of a ConMap CED mask for one sample point of the real-world dataset, which represents the regional geomorphic signature. Red indicates positive and blue indicates negative CED values. The coordinates are in meters. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

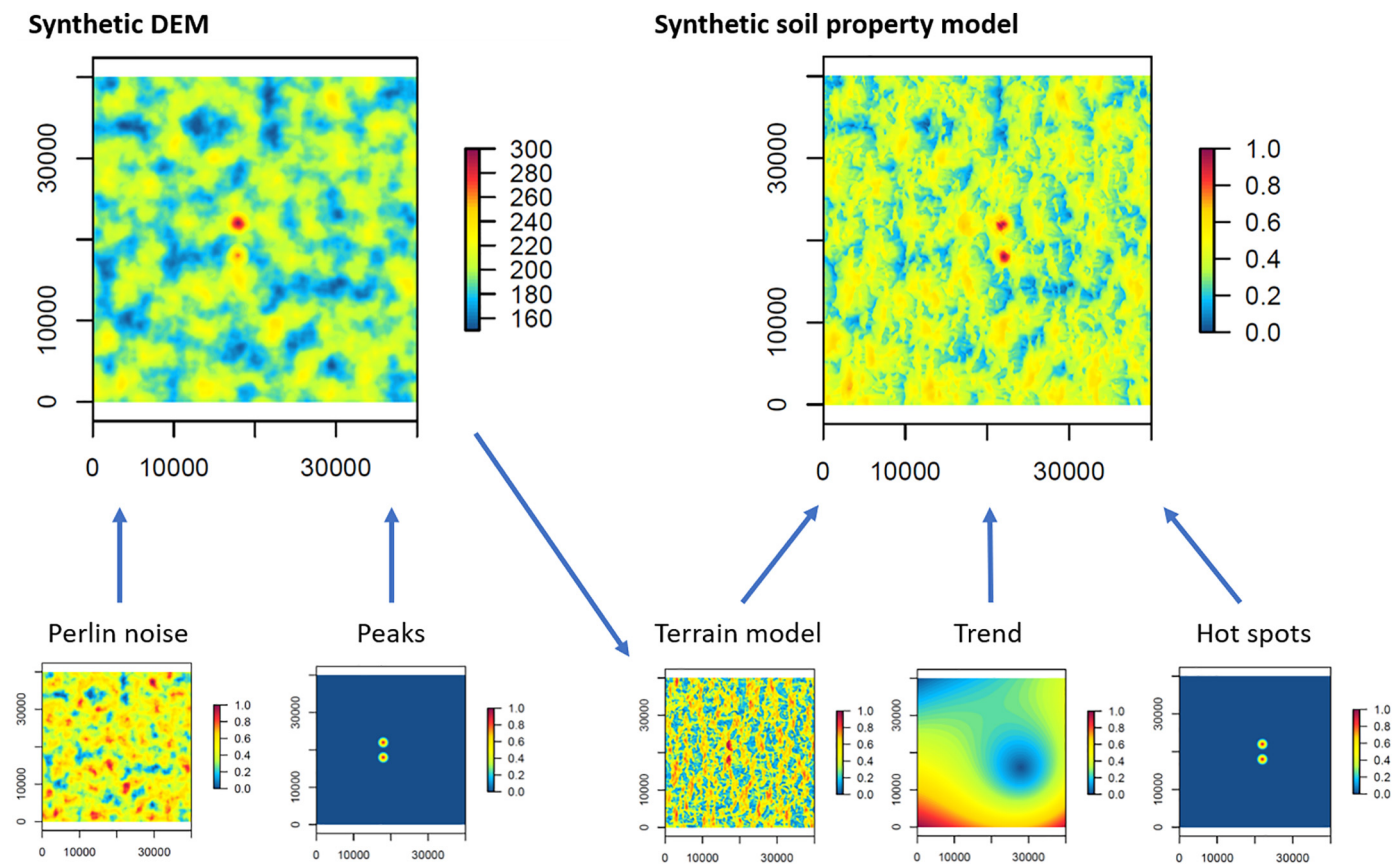


Fig. 5. Components of the synthetic DEM and the synthetic soil property map.

2.2. Datasets

2.2.1. Synthetic data

We created a synthetic data set with specific surface characteristics (peaks) and an associated artificial soil property distribution including hot spots. The DEM was generated using Perlin noise (Perlin, 1985, 2002; Pedersen and Peck, 2018) with two cone-shaped peaks imposed west of the center. The soil data was generated by a simple model consisting of a combination of elevation, slope, eastness, a spatial trend model with two hotspots imposed east of the center (Fig. 5).

The hypothetical teleconnected soil formation scenario is that of material (e.g. heavy metals) transported by wind from west to east. If the air-flow was disturbed by a surface feature (i.e. the two peaks), part of the material would be deposited in the lee of the peaks and create hot spots there. The idea is to see if we can identify these peaks as key controls for the hotspots.

A conditioned Latin Hypercube Sampling (cLHS) sample set (Minasny and McBratney, 2006; Roudier, 2018) consisting of 300 samples was drawn from a spatial subset of the synthetic soil property model (Fig. 6). The subset was chosen so that the controls, i.e. the peaks, were not included, but the soil hotspots were.

2.2.2. Rhine-Hesse data

The data set from Rhine-Hesse, Germany (Fig. 7) is an example of a complex teleconnected system that leads to a strongly autocorrelated distribution of soil properties (Behrens et al., 2010, 2018c). The formation of the silt content of the topsoil is related to wind erosion and loess translocation from the Rhine-Main lowlands to the surrounding heights of Rhine-Hesse. The teleconnected system is larger than the extent of the actual data set. Therefore, it may not be possible to decode the system completely. However, we use this dataset because we know that soil formation is based on an identifiable, long-range

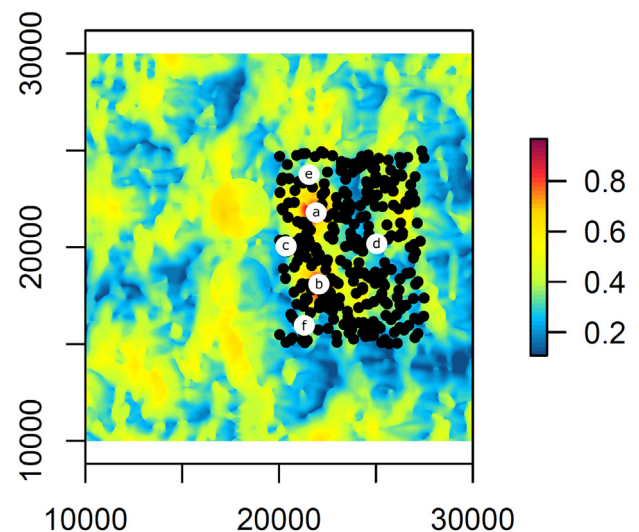


Fig. 6. 300 cLHS samples based on the synthetic soil property model. The coordinates are in meters. The highlighted samples (a–f) correspond to the samples shown in Fig. 9.

teleconnection. It is the same dataset used in the original presentation of ConMap (Behrens et al., 2010), and other contextual mapping (Behrens et al., 2014, 2018a, 2018b).

The average annual precipitation of this loess region ranges from 500 to 550 mm, rising to 850 mm in the surrounding mountain regions. Three hundred and forty-two samples with measurements of topsoil silt content (5–82%) for the 0–10 cm depth interval were available (Fig. 7). The resolution of the DEM was set at 40 m.

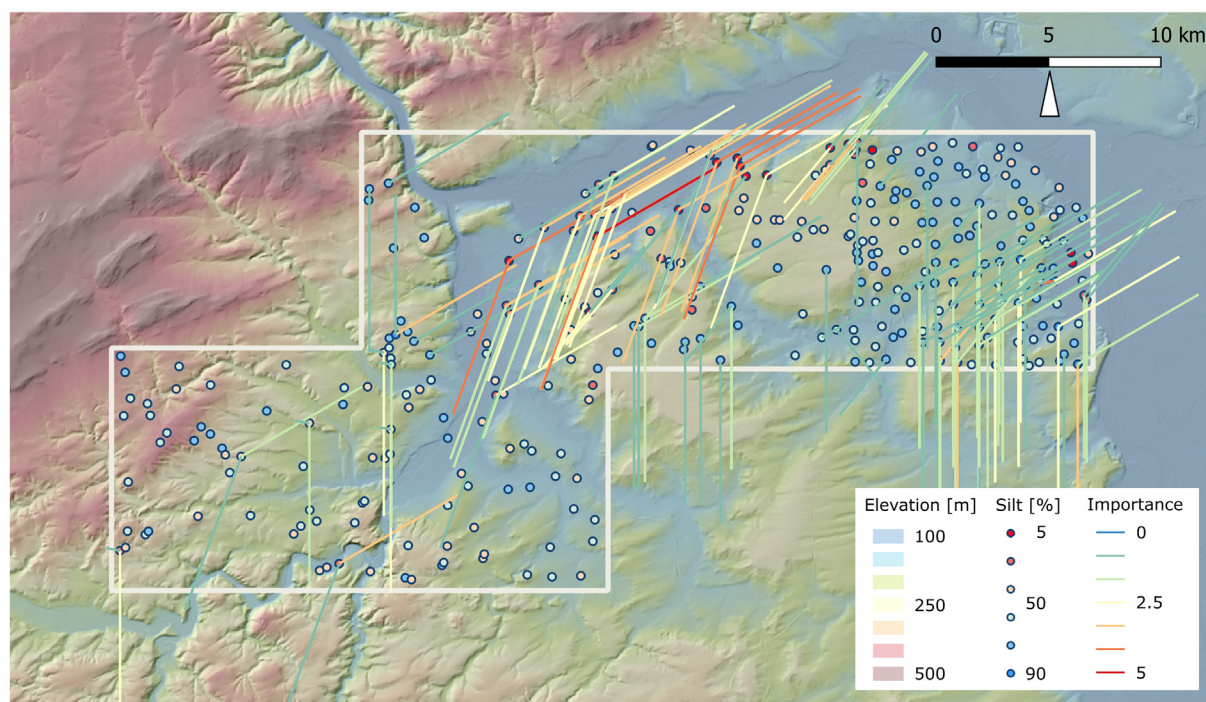


Fig. 7. Rhine-Hesse dataset. The map shows the sample locations ($n = 342$) as well as the teleconnection systems for those samples with the highest local feature importances draped over the DEM. The lines, representing the teleconnections, go from the sample locations to the site of the CED location with the highest importance. For each sample the two highest connections are shown to visualize the entire teleconnection system.

2.3. Visualizing teleconnections

In general, the importance of CEDs can be visualized in different ways. One way is to visualize the ConMap mask as a series of points or circles originating from the center of the mask (as shown in Fig. 4). The importance of the CEDs can then be represented by varying the color density of the points, with darker points indicating more important CEDs (as shown in Fig. 8). Alternatively, the diameter of the circles can be varied proportionally, with larger circles showing more important CEDs (as shown in Fig. 9). Finally, you can draw lines that connect the major CEDs to the center where they exert a strong influence, and color these lines to indicate their relative importance (as in Fig. 7).

All approaches can be used to visualize the local and overall importance of the features. While the overall feature importance has to be

visualized with one mask for the entire sample set, the local feature importance has to be visualized separately for each sample. For the latter, the intent is to identify and illustrate the most important CEDs that surround individual samples of interest.

To visualize the local importance of the CEDs for the synthetic data set, and thus to be able to unravel the teleconnected system, we used proportional circles. We selected sample locations on, between and around the soil property hotspots, to optimally visualize the teleconnection between the peaks in the DEM and the soil property hotspots.

For the real-world dataset we used directional lines from the sample location to the first two most important CEDs to visualize the teleconnected system based on local feature importance. Therefore, we used all samples above a local feature importance value of 1 [percent

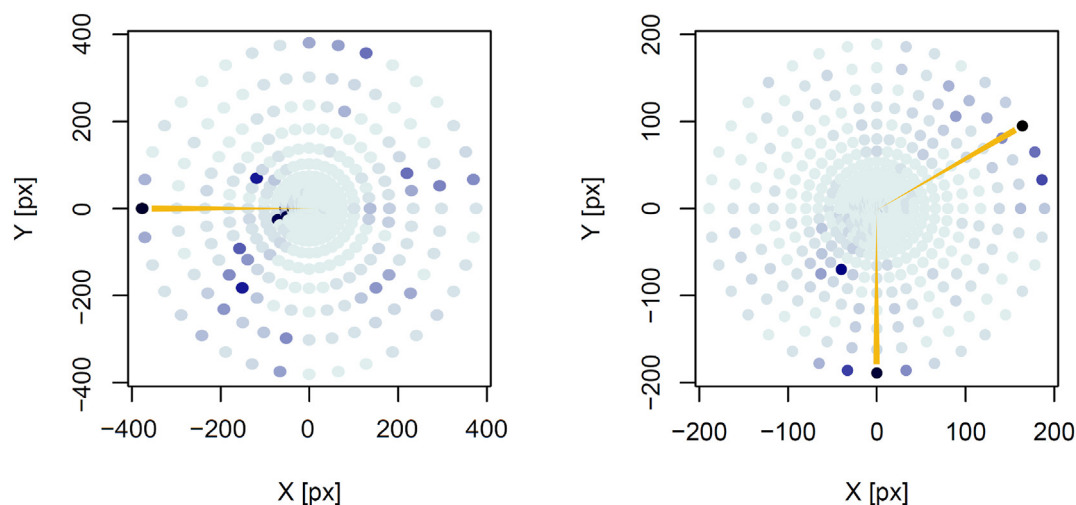


Fig. 8. Overall feature importance of the CED locations for the synthetic (left) and the Rhine-Hesse dataset (right). Darker dots represent higher importance values. The orange lines represent the direction from the center to the CED(s) with the highest importance. The coordinates are in pixels.

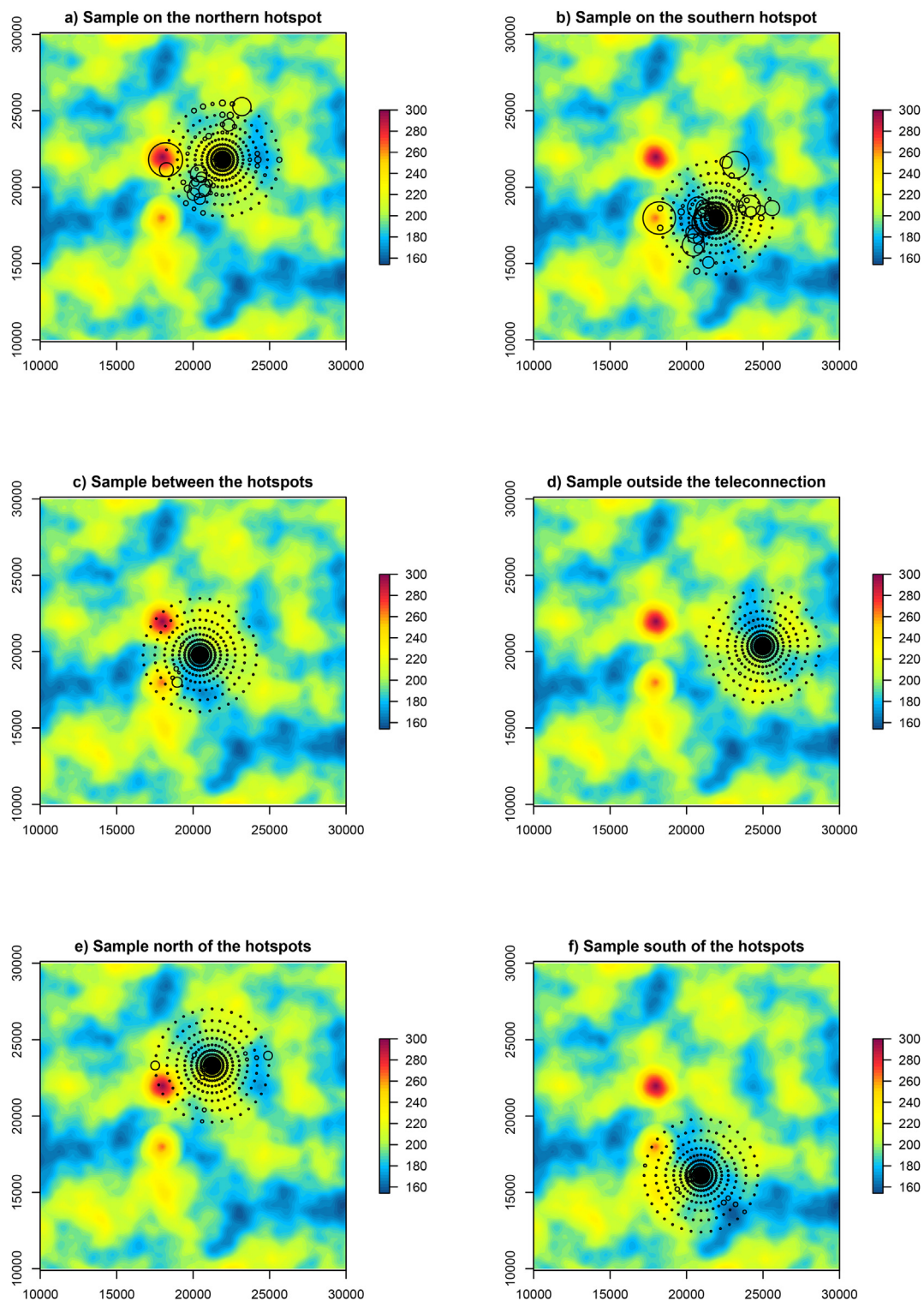


Fig. 9. Local feature importance of the CEDs for six selected sample locations (Fig. 6) in the synthetic dataset, draped over the DEM. The brighter red areas are elevated peaks in the DEM which we assumed to influence the soil property distribution at the hotspots of the synthetic soil property distribution (Fig. 5). The larger the circles, the higher the local importance of the CEDs for the specific cell at the center of the ConMap mask centered over that specific sample point. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

increase in mean square error Breiman (2001)] simultaneously to visualize the teleconnected system.

3. Results

3.1. Synthetic data

Fig. 8a shows the overall feature importance of the CEDs for the synthetic data set. The darker the dots, the greater the importance. It

turns out that only a few CEDs have a relatively high overall feature importance. Since the CEDs represent specific relative coordinates in relation to each sample location, the local feature importance can be mapped to visualize the teleconnections.

Fig. 9 shows the local importance for six different sample locations, which were selected to visualize the locally most important terrain features for different parts of the area (Fig. 6). The samples are located either directly on the main soil property hotspots or in the immediate surroundings of the main hotspots. Although five out of six of the selected samples exhibit CEDs which overlap the peaks, only the two samples located directly on top of the hotspots reveal the peaks as principal controls of the distribution of the soil property of interest. For these specific “hot spot” locations, the largest circles are centered over the synthetic elevation peaks and therefore identify these peaks as the most likely principal controls on downwind soil formation. This demonstrates that ConMap can be used to decipher teleconnected soil environmental response systems. All other sample locations (not centered over hot spots) exhibit mostly local influences based on the general soil property genesis, which is mainly based on local process systems (cf. Section 1). All samples also show some higher importance values (larger CED importance circles) at larger radii/scales. This is due to the spatial trend model, which is part of the synthetic soil property distribution (Fig. 5).

The general orientation of the teleconnected system can be determined by analyzing the overall feature importance and by drawing a line from the center of the mask to the feature with the highest importance (Fig. 8). The synthetic example shows an east-west connection from the sample locations to the most important CED, indicating that the CED located to the west has a strong influence on (some) samples in the east.

3.2. Rhine-Hesse data

Fig. 8b shows the overall feature importance of the CEDs for the Rhine-Hesse data set. As with the synthetic data set, only a few CEDs have high overall importance values. In contrast to the synthetic example, we deemed it important to analyze and interpret all important features in the Rhine Hesse area because the relevant landscape features are larger. It can be seen that the teleconnected system has two directional components relative to the sample locations: a north-south orientation and a southwest-northeast orientation. The north-south axis reveals the general connection to the loess plateaus in the south. The southwest-northeast direction reveals the influence of the valley and also reflects the angle of the main valley in the study area.

The most important CEDs, based on the local feature importance analysis, are shown in Fig. 7. The second most important feature is also displayed to visualize the complex (and locally varying) teleconnected system over the entire study area. The most important teleconnected system spans from the main valley to the plateaus in the south (Fig. 7). Hence, both the valley from which the silt was eroded and removed by wind and the plateau on which the silt was deposited and accumulated are important indicators for the teleconnected process system of silt translocation.

4. Discussion

ConMap can help identify and unravel teleconnected systems but it cannot entirely reveal the actual process or control. In our real-world example, the main control of the spatial variation in silt content is the complex interaction between terrain and climate. However, interpretations must be made with care because one cannot be sure that the “detected” system is a proxy for some other system(s). We suggest that further research is needed on the analysis and interpretation of teleconnected systems. As we have shown here, the advantage of using ConMap is that the controls, or proxies of the controls, of teleconnected systems can be spatially located. This is because ConMap

simultaneously provides relevant information on variation in relative topography and spatial location across multiple spatial extents, scales and directions. This type of analysis of teleconnected systems cannot be easily achieved using other multiscale methods such as Gaussian pyramids (Behrens et al., 2018a, 2018b), wavelets (Lark and Webster, 2004; Biswas et al., 2013) or convolutional neural networks (Padarian et al., 2019). Although an advantage of the Gaussian pyramids over convolutional neural networks is that the scaled covariates can be interpreted directly because they have a known meaning (Behrens et al., 2018b). Both approaches, however, do not support extraction of the position of an influencing landscape element, as they both spatially convolve the original input data with types of low-pass filters. Combining the interpretability of the Gaussian pyramid mixed scaling covariates and ConMap's ability to reveal teleconnected systems might be a way forward.

In addition to providing information on topographic spatial context, the CEDs used as explanatory covariates here may also be acting as measures of relative spatial location, in a manner similar to that achieved by other explicit measures of spatial location such as Euclidian Distance Fields (Behrens et al., 2018c). In this sense, these locational covariates (CEDs) may act to achieve an effect similar to the way weights are assigned to values at known locations when interpolating using geostatistical methods. Hence, the locational components of ConMap covariates do offer some explanatory power, even in cases of subdued landscapes where there are no easily identifiable relationships between terrain derivatives and properties or classes to be predicted. This also helps to partially explain why the residuals do not exhibit spatial structure and so do not benefit from a separate kriging step as implemented in regression-kriging (cf. Behrens et al., 2018a).

Methodologically, it is important to consider the relevant range of scales. In general, we see that if coarser scales are included in a model, the prediction accuracy increases to a certain point or spatial range beyond which prediction accuracy does not increase any further (Behrens et al., 2010, 2014, 2018a). This maximum relevant spatial range should be used as a basis for the analysis of teleconnections. If, however, more coarse scales are included, some false relationships could be created which may then undermine the interpretation. If the maximum radius is set too small, relevant information needed to resolve the teleconnected system may be missing.

5. Conclusions

We have defined the three most important soil environmental response systems and shown that multiscale modelling approaches make it possible to decode the 3rd process system, i.e. that of teleconnected environmental systems. This offers additional tools for the analysis, description and interpretation of the ‘unknown’ part of the deterministic component of the universal model of spatial variation, commonly expressed as spatial dependence. If the three soil environmental response systems are seen as effects of scale and short-range to long-range interactions of structural covariates, they can be integrated in Jenny's CLORPT concept without the need to explicitly account for spatial dependency.

As with any other modelling approach, there are a number of limitations. For example, if the current terrain surface does not affect the spatial distribution of soils, an approach based on elevation differences alone may not work well. Of course, other covariates, such as gamma radiometrics, if available, can also be used to extract ConMap type derivatives instead of, or in combination with, the DEM.

Our approach here offers indications to identify possibly teleconnected systems, but no detailed explanation of causality. For detailed analysis and interpretation, future studies should apply this approach together with the inverse approach, i.e. mapping the region influenced by a feature. Domain knowledge is always required to interpret relationships and contexts. The relevant covariates and the final modelling results should be interpreted in the same way as in more

classical pedology. Our approach is a first step towards a more holistic system for interpreting spatial and environmental controls on soil formation.

Acknowledgements

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